

ECE148: Signal Processing Applications

Assignment 1: Signal Sampling and Aliasing (Part A)

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1 Orthogonal Decomposition

Given a low-pass signal $f(t)$ bandlimited to $[-\frac{\omega_0}{2}, \frac{\omega_0}{2}]$, sampled with spacing Δt , producing samples $f(n\Delta t)$.

Fourier Representation

The Fourier transform is:

$$F(j\omega) = \sum_{n=-\infty}^{\infty} f_n e^{-j\omega n\Delta t}$$

with coefficients:

$$f_n = \frac{1}{\omega_0} \int_{-\frac{\omega_0}{2}}^{\frac{\omega_0}{2}} F(j\omega) e^{j\omega n\Delta t} d\omega$$

The inverse Fourier transform is:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega$$

Since $f(t)$ is bandlimited:

$$f(n\Delta t) = \frac{1}{2\pi} \int_{-\frac{\omega_0}{2}}^{\frac{\omega_0}{2}} F(j\omega) e^{j\omega n\Delta t} d\omega$$

Comparing expressions, we obtain:

$$f_n = \frac{2\pi}{\omega_0} f(n\Delta t) = \Delta t f(n\Delta t)$$

Substituting into the inverse Fourier transform:

$$f(t) = \frac{1}{2\pi} \int_{-\frac{\omega_0}{2}}^{\frac{\omega_0}{2}} \left[\sum_{n=-\infty}^{\infty} \Delta t f(n\Delta t) e^{-j\omega n\Delta t} \right] e^{j\omega t} d\omega$$

$$= \sum_{n=-\infty}^{\infty} f(n\Delta t) \frac{\Delta t}{2\pi} \int_{-\frac{\omega_0}{2}}^{\frac{\omega_0}{2}} e^{j\omega(t-n\Delta t)} d\omega$$

Evaluating the integral:

$$\frac{\Delta t}{2\pi} \int_{-\frac{\omega_0}{2}}^{\frac{\omega_0}{2}} e^{j\omega(t-n\Delta t)} d\omega = \text{sinc}\left(\frac{\omega_0}{2}(t-n\Delta t)\right)$$

Thus:

$$f(t) = \sum_{n=-\infty}^{\infty} f(n\Delta t) \psi_n(t)$$

where:

$$\psi_n(t) = \text{sinc}\left(\frac{\omega_0}{2}(t-n\Delta t)\right)$$

Orthogonality of Basis Functions

Consider:

$$\langle \psi_n, \psi_m \rangle = \int_{-\infty}^{\infty} \psi_n(t) \psi_m^*(t) dt$$

Using Parseval's theorem:

$$\langle \psi_n, \psi_m \rangle = \frac{\Delta t^2}{2\pi} \int_{-\frac{\omega_0}{2}}^{\frac{\omega_0}{2}} e^{-j\omega(m-n)\Delta t} d\omega$$

Evaluating:

$$= \Delta t \text{sinc}\left(\frac{\omega_0}{2}(m-n)\Delta t\right)$$

Using the sampling condition $\omega_0 \Delta t = 2\pi$:

$$= \Delta t \text{sinc}(\pi(m-n))$$

Therefore:

$$\langle \psi_n, \psi_m \rangle = \begin{cases} \Delta t, & m = n \\ 0, & m \neq n \end{cases}$$

2 Application of Aliasing: Time-Domain Partition

The time-domain function

$$f(t) = \sum_{n=1}^N f_n(t)$$

is a low-pass signal partitioned into N sub-components. The frequency spectra of $f(t)$ and all sub-components $f_n(t)$ span from $-\frac{\omega_0}{2}$ to $\frac{\omega_0}{2}$, giving a total bandwidth of $B = \omega_0$.

Modulating each of the N sub-components with an extra frequency term $e^{jk_n\omega_0 t}$ yields:

$$\hat{f}(t) = \sum_{n=1}^N f_n(t) e^{jk_n\omega_0 t}$$

In the frequency domain, this corresponds to a shift:

$$\hat{F}(j\omega) = \sum_{n=1}^N F_n(j(\omega - k_n\omega_0))$$

After sampling $\hat{f}(t)$ with spacing Δt (sampling frequency $\omega_0 = \frac{2\pi}{\Delta t}$), the spectrum becomes:

$$\hat{F}_s(j\omega) = \sum_{n=1}^N \sum_{m=-\infty}^{\infty} F_n(j(\omega - (k_n + m)\omega_0))$$

Passing the sampled signal through an ideal low-pass filter with cutoff $|\omega| \leq \frac{\omega_0}{2}$ selects only the terms where:

$$\omega - (k_n + m)\omega_0 \in \left[-\frac{\omega_0}{2}, \frac{\omega_0}{2} \right]$$

which is only possible when $k_n + m = 0$, resulting in $m = -k_n$. Therefore, all other aliases are removed, and the spectrum collapses to:

$$\hat{F}_{LP}(j\omega) = \sum_{n=1}^N F_n(j\omega) = F(j\omega)$$

Reconstructing the signal returns the original unmodulated signal:

$$\hat{f}_{out}(t) = f(t)$$

The original signal $f(t)$ is perfectly recovered. The integer frequency shifts $k_n\omega_0$ are completely removed by aliasing and low-pass filtering.

3 Application of Aliasing: Frequency-Domain Partition

Consider a low-pass periodic function $g(t)$, which can be represented by its Fourier series expansion:

$$g(t) = \sum_{n=-N}^N G_n e^{jn\omega_x t}$$

where ω_x is the fundamental frequency and G_n are the Fourier coefficients. The frequency spectrum of $g(t)$ spans:

$$[-N\omega_x, +N\omega_x]$$

giving a total bandwidth $B = 2N\omega_x < \omega_0$.

We now modulate each of the $2N + 1$ Fourier components with an extra single-frequency term $e^{jk_n\omega_0 t}$, forming:

$$\hat{g}(t) = \sum_{n=-N}^N G_n e^{j(n\omega_x + k_n\omega_0)t}$$

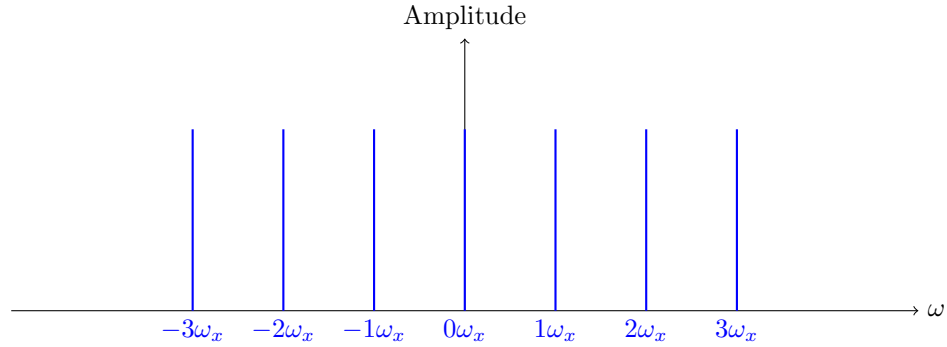
where k_n are arbitrary integers.

(a) Fourier Spectrum of $g(t)$

The Fourier spectrum of $g(t)$ consists of discrete impulses located at

$$\omega = n\omega_x, \quad n = -N, -N + 1, \dots, N - 1, N,$$

with amplitudes G_n .



(b) Fourier Spectrum of $\hat{g}(t)$

Modulating each component by $e^{jk_n\omega_0 t}$ shifts each impulse:

$$\omega = n\omega_x + k_n\omega_0, \quad n = -N, \dots, N$$

The spectrum remains discrete, but each spike is now shifted by an integer multiple of ω_0 .

(c) Sampling and Low-Pass Filtering

Sampling $\hat{g}(t)$ with spacing Δt ($\omega_s = 2\pi/\Delta t = \omega_0$) produces periodic copies of the spectrum in frequency, spaced by ω_0 :

$$\hat{G}_s(j\omega) = \sum_{m=-\infty}^{\infty} \sum_{n=-N}^N G_n \delta(\omega - (n\omega_x + k_n\omega_0) - m\omega_0)$$

Passing this through an ideal low-pass filter with cutoff $|\omega| \leq \omega_0/2$ keeps only the components that satisfy

$$n\omega_x + k_n\omega_0 + m\omega_0 \in \left[-\frac{\omega_0}{2}, \frac{\omega_0}{2}\right]$$

Since $B = 2N\omega_x < \omega_0$, each $n\omega_x$ lies within $[-\omega_0/2, \omega_0/2]$ after choosing $m = -k_n$. Therefore, all frequency shifts by $k_n\omega_0$ are folded back into the baseband via aliasing. The low-pass filter output spectrum becomes:

$$\hat{G}_{LP}(j\omega) = \sum_{n=-N}^N G_n \delta(\omega - n\omega_x) = G(j\omega)$$